

# Experimental Design

Five controlled sweeps — isolating threshold, judge, and architecture effects

| Sweep File        | Purpose                 | Conditions | Threshold | Generator / Judge      |
|-------------------|-------------------------|------------|-----------|------------------------|
| sweep_phase1      | Phase 1 Baseline        | A / B / C  | 0.70      | qwen3-32b / qwen3-32b  |
| sweep_thresh85_b  | Threshold Sensitivity B | B only     | 0.85      | qwen3-32b / qwen3-32b  |
| sweep_thresh85_c  | Threshold Sensitivity C | C only     | 0.85      | qwen3-32b / qwen3-32b  |
| sweep_judge_sep_a | Judge Separation A      | A only     | —         | qwen2.5-7b / qwen3-32b |
| sweep_judge_sep_b | Judge Separation B      | B only     | 0.85      | qwen2.5-7b / qwen3-32b |

All sweeps: 19 SWE-bench Lite instances across django, sympy, scikit-learn, astropy

# Phase 1 Results — Headline Numbers

19 instances × 3 conditions — qwen3-32b generator and judge

0.878

A mean score  
( $\sigma = 0.145$ , min 0.30)

0.881

B @ 0.70 mean  
( $\sigma = 0.069$ , min 0.72)

0.926

B @ 0.85 mean  
( $\sigma = 0.044$ , min 0.85)

0.887

C @ 0.70 mean  
( $\sigma = 0.126$ , min 0.40)

0.721

C @ 0.85 mean  
( $\sigma = 0.353$ , min 0.00)

## ▲ B@0.85 beats B@0.70

Mean +0.045, stdev halved (0.069→0.044), hard floor at 0.85. Raising the threshold forces more refine iterations and eliminates all below-threshold answers.

## ✓ Reviewer narrows variance

A spans 0.30–0.96; B@0.85 spans 0.85–1.00. The refine loop is effectively a score floor: no instance escapes without satisfying the rubric.

## ⚠ Condition C is bimodal at 0.85

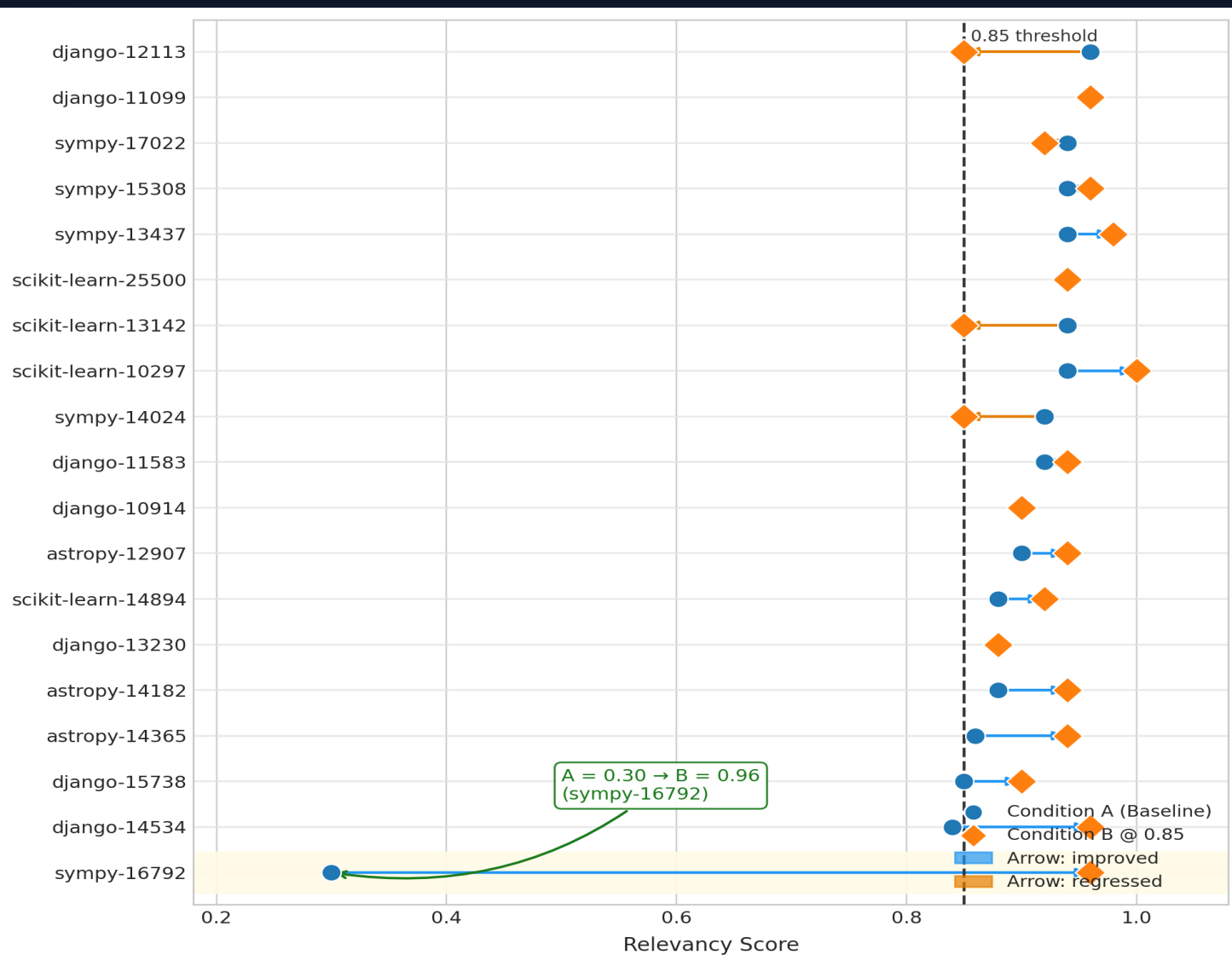
14 / 19 instances score 0.86–1.00. The remaining 5 regress sharply ( $\leq 0.40$ ), including one at 0.00. Aggressive refinement with an action agent is not uniform improvement.

## ★ Judge separation amplifies lift

With a weaker Ollama 7B generator the Reviewer adds  $\Delta = +0.114$  vs only  $\Delta = +0.003$  when the generator is strong — the Reviewer matters most where it is needed most.

# Finding 1 — Reviewer Enforces a Hard Score Floor

Every B@0.85 instance  $\geq 0.85 \cdot$  Baseline A spans 0.30  $\rightarrow$  0.96



0.30  $\rightarrow$  0.96

sympy-16792: biggest single lift.  
Baseline scored 0.30 — Reviewer drove it to 0.96.

$\sigma$  halved

A stdev = 0.145  $\cdot$  B@0.85 stdev = 0.044.  
Refine loop compresses the score distribution.

Hard minimum

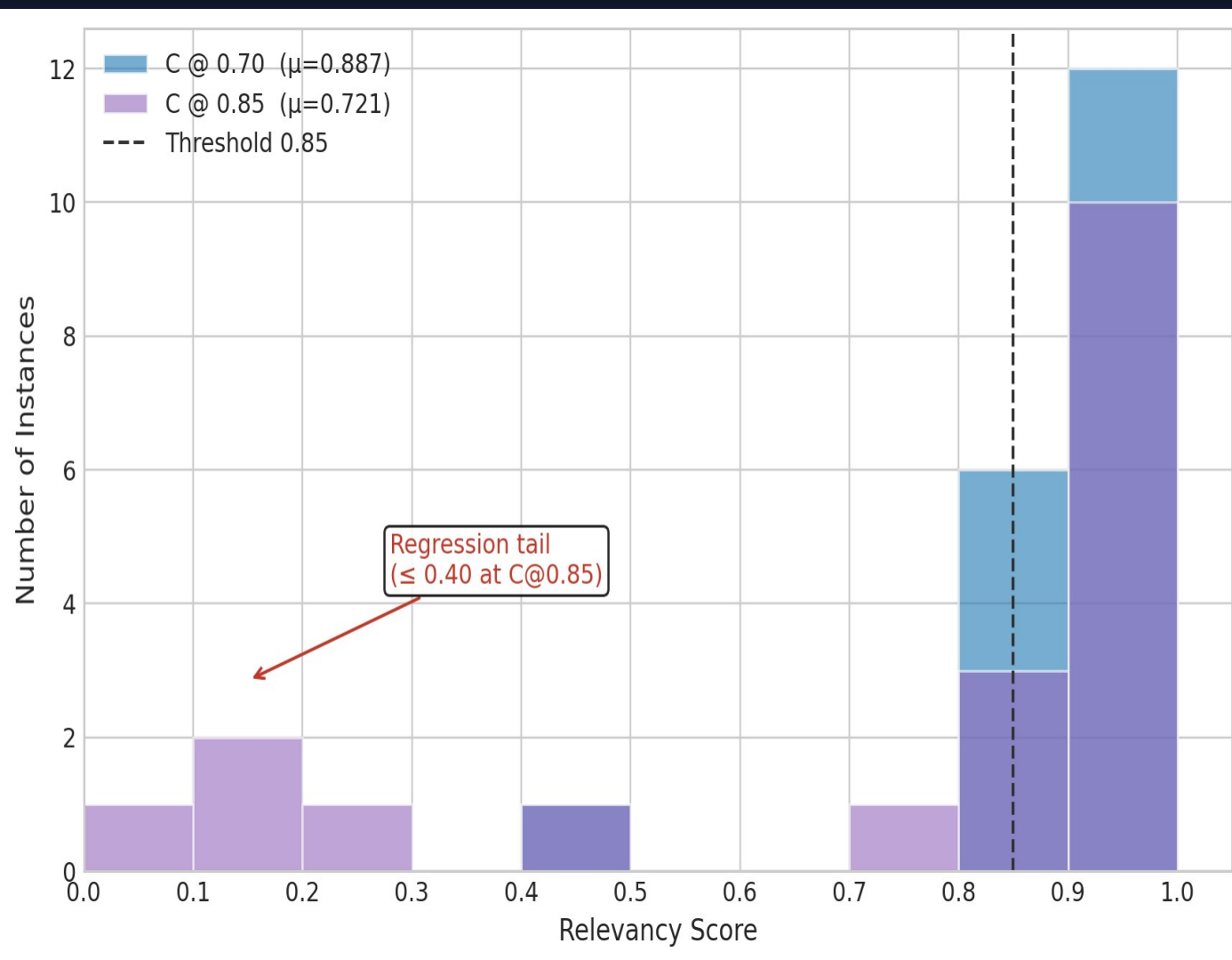
B@0.85 minimum = 0.85 across all 19 instances.  
The rubric + refine acts as a guaranteed floor.

73 % of instances lifted

14 of 19 instances improved under B vs A.  
Only 5 showed marginal regression (all  $\geq 0.85$ ).

# Finding 2 — Condition C is Bimodal at Threshold 0.85

Aggressive refinement with an action agent is not uniformly beneficial



## High cluster

14 / 19 instances

Score 0.86 – 1.00

Action agent succeeds reliably

## Regression tail

5 / 19 instances

Score  $\leq 0.40$  (one at 0.00)

Aggressive refine degrades output

## Root cause

### Context explosion

OpenHands accumulates ~150K tokens by step 30 — judge loses coherence

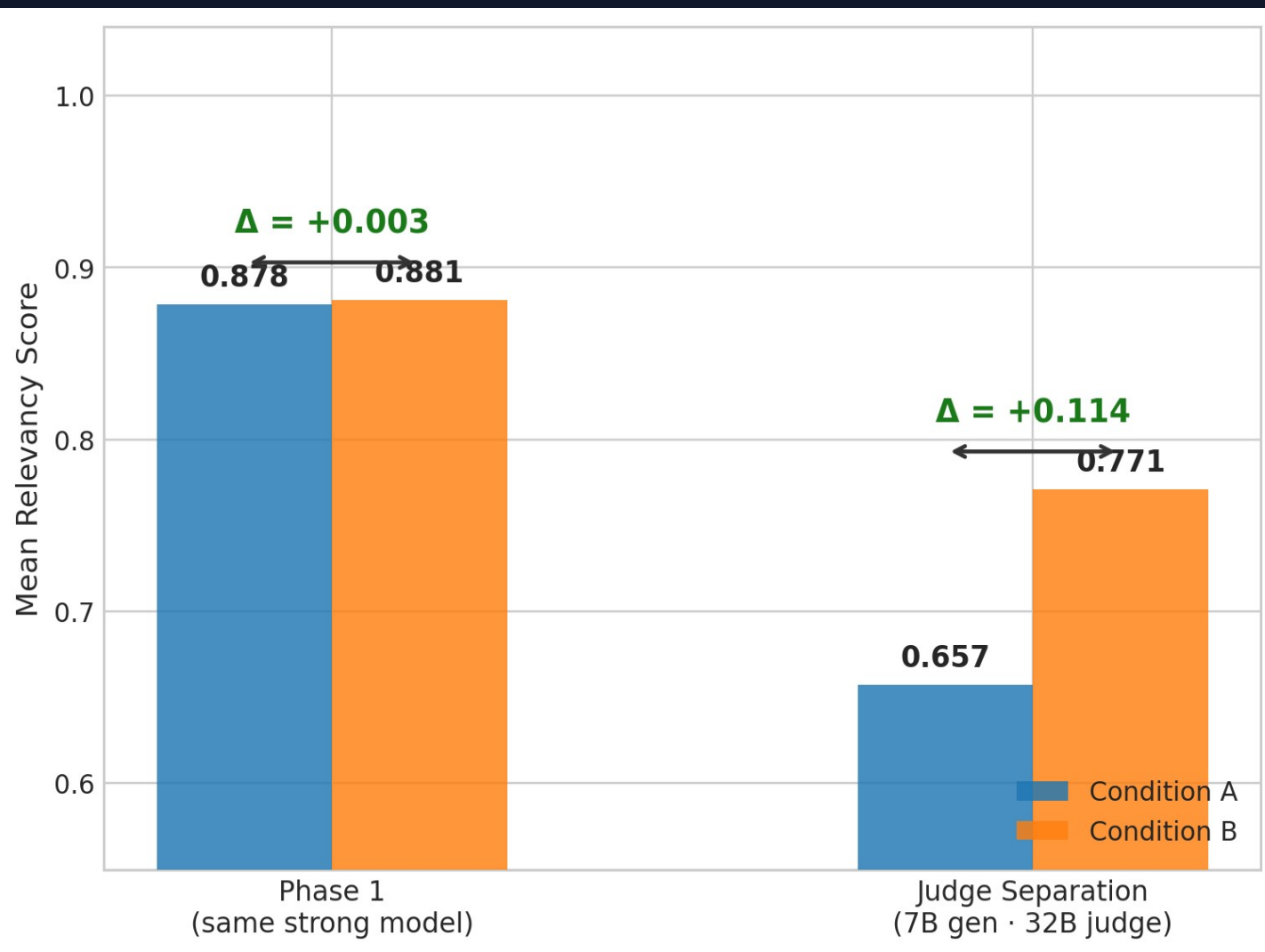
## Implication

### Threshold matters

C@0.70 mean = 0.887 vs C@0.85 = 0.721  
Higher threshold amplifies tail risk

# Finding 3 — Reviewer Adds Most Value When Generator is Weak

Judge separation experiment: Ollama qwen2.5-7B generator · qwen3-32B judge



## $\Delta = +0.114$ (weak gen)

Mean A = 0.657, mean B = 0.771.  
When the generator struggles the Reviewer's refine loop delivers the largest lift.

## $\Delta = +0.003$ (strong gen)

Mean A = 0.878, mean B = 0.881.  
A capable generator leaves little room for the Reviewer to improve.

## Architectural implication

The Reviewer is not redundant — it is a safety net that scales with generator weakness. Deploy it whenever the generator is uncertain.

# Conclusions, Limitations & Future Work

## CONCLUSIONS

### Rubric-based gating works

B@0.85 lifts mean score by +0.048 over A and enforces a hard minimum — no instance escapes below 0.85.

### Reviewer value scales with generator weakness

$\Delta = +0.003$  (strong model) vs  $\Delta = +0.114$  (7B model). The Reviewer is most valuable where the generator is most uncertain.

### Hallucination Bridge generalises the rubric

The same 5-dimension rubric audits OpenHands action traces — a structurally distinct hallucination class becomes detectable without fine-tuning.

### Bimodality is a real risk at C@0.85

Aggressive refinement with an autonomous action agent can degrade output when the judge loses coherence under large context.

## LIMITATIONS

- Absolute scores are LLM-regime sensitive — only within-regime ordering is comparable.
- Free-tier throttling affected Condition C; token costs per run are high (~150K at step 30).
- 19-instance set provides directional signal but limited statistical power.

## FUTURE WORK

- Full 50-instance SWE-bench Lite sweep for statistical significance.
- Formal per-action hallucination metric on OpenHands traces.
- Threshold calibration — per-domain adaptive thresholding.
- Evaluate with stronger open-source models (Qwen2.5-72B, GPT-4o).

# THANK YOU

## FOR YOUR TIME

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*Development of a Lightweight Hallucination Detection Algorithm for the Reviewer Agent in a Code-Generation Multi-Agent System*

19

SWE-bench instances

5

Sweep configurations

3

Architectural conditions

+0.114

Max Reviewer lift

0.85

B@0.85 score floor